

PHYS 3200 - Fall 2026

End of Topic Quiz 3

September 25, 2026

Name and ID #:

Instructions:

1. You may use:
 - A calculator.
 - Notes.
 - Writing utensil: pencil and eraser are best.
2. If you have a question about a problem (confused about the situation, need some missing piece of information, etc.), please raise your hand and ask!
3.

 for each part.
4. Credit will not be given if your answers are too messy or obscure to read. Remember: a grader who isn't squinting and spending extra time trying to decipher mysterious scratches is a happy grader.
5. This quiz must be turned in by the end of the class period.

1. A rocket travels in the positive x -direction through force-free space. At some time, the rocket has mass m and velocity $\dot{x}$. The rocket ejects fuel backward with a constant exhaust speed v_{ex} relative to the rocket.

During a short time interval, the rocket's mass changes by dm , where $dm < 0$, and its velocity changes by $d\dot{x}$. The positive mass of the expelled fuel is therefore $-dm$.

- (a.) Write the total momentum immediately before and immediately after the fuel is ejected. Neglect products of small quantities, such as $dm d\dot{x}$, and use conservation of momentum to show that

$$m d\dot{x} = -v_{\text{ex}} dm.$$

- (b.) The rocket initially has mass m_0 and velocity $\dot{x}_0$. Separate and integrate the equation from part (a) to find the rocket's velocity $\dot{x}$ when its mass has decreased to m .

- (c.) Find the change in the rocket's velocity when its mass decreases from m_0 to $m_0/2$.

2. Consider two particles, each with mass m . At $t = 0$, their positions and velocities are

$$\mathbf{r}_1(0) = -a\hat{\mathbf{x}}, \quad \dot{\mathbf{r}}_1(0) = v_0\hat{\mathbf{y}},$$

$$\mathbf{r}_2(0) = a\hat{\mathbf{x}}, \quad \dot{\mathbf{r}}_2(0) = v_0\hat{\mathbf{x}}.$$

The external forces acting on the particles are

$$\mathbf{F}_1 = 0, \quad \mathbf{F}_2 = F_0\hat{\mathbf{x}},$$

where F_0 is a positive constant.

(a.) Find the initial position $\mathbf{R}(0)$ and initial velocity $\dot{\mathbf{R}}(0)$ of the center of mass.

(b.) Find the acceleration $\ddot{\mathbf{R}}(t)$ of the center of mass.

(c.) Find the velocity $\dot{\mathbf{R}}(t)$ and position $\mathbf{R}(t)$ of the center of mass.